



Deepfake Detection Using Hand-Crafted Features and Deep Learning

Submitted by :

Soumadip Singha Mahapatra

Roll No- 2022ITB097

B.Tech, Information Technology

IIEST, Shibpur

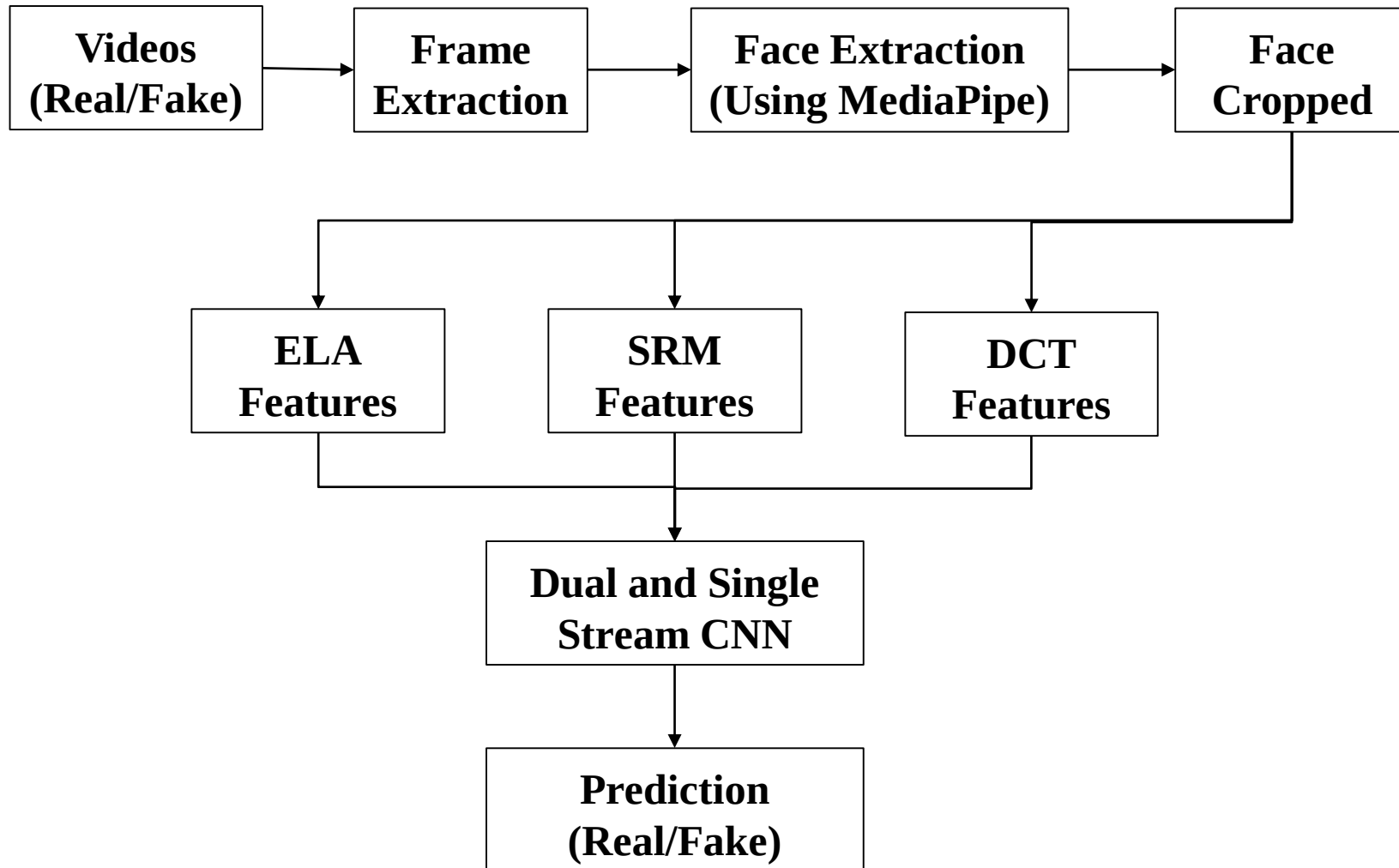


Problem Statement and Motivation

- ❑ Deepfake technology is rapidly advancing, creating highly realistic synthetic media that threatens digital trust.**
- ❑ Major risks include political misinformation, identity theft, financial fraud, and manipulation of legal evidence.**
- ❑ A robust multi-modal detection system is required, combining classical forensic features with deep learning and achieving 85–95% accuracy to avoid harmful false positives and false negatives.**



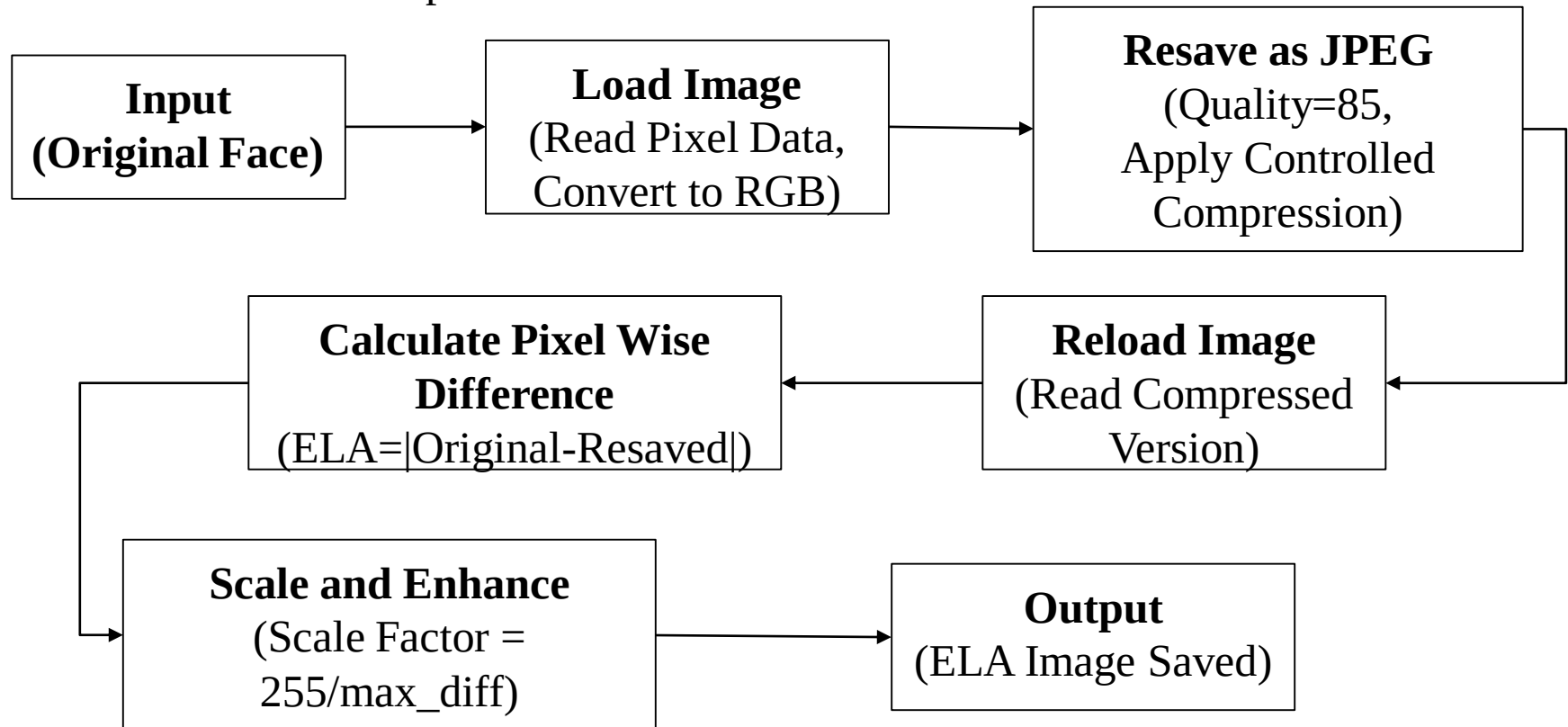
Complete Pipeline Overview



Hand-Crafted Feature Extraction

❑ Error Level Analysis(ELA)

Error Level Analysis detects image manipulation by analyzing JPEG compression artifacts.



Hand-Crafted Feature Extraction

❑ Error Level Analysis(ELA)

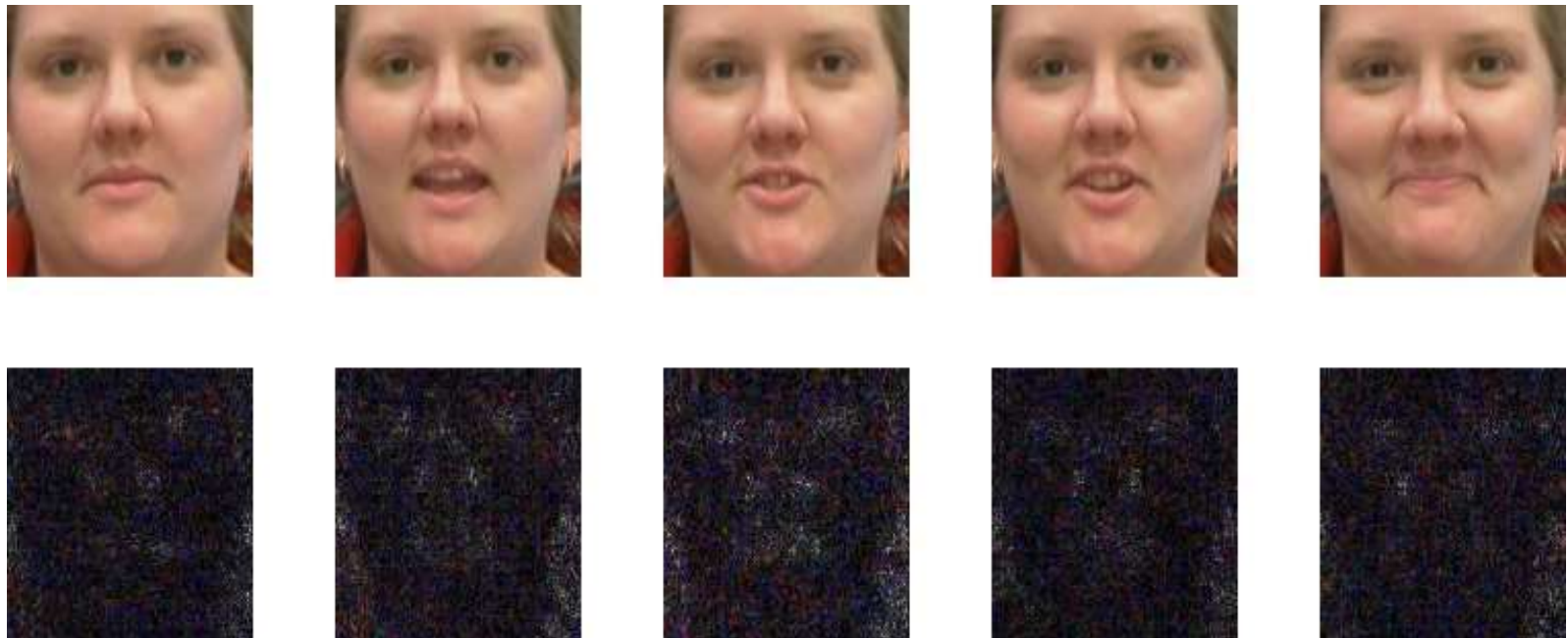
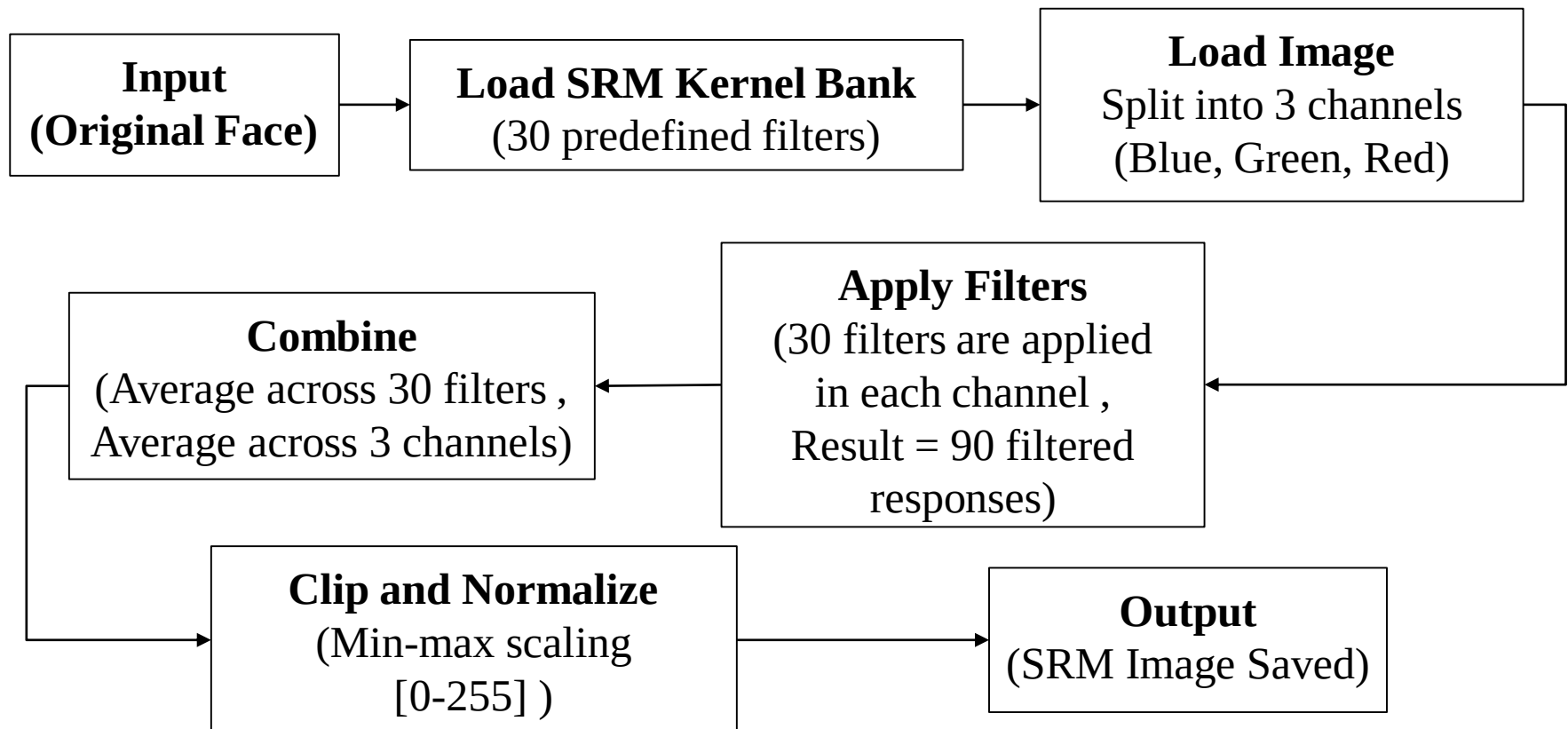


Fig: Authentic Image and Its ELA

Hand-Crafted Feature Extraction

❑ Noise Residuals from Spatial Rich Model (SRM)

SRM extracts noise residuals from images using a fixed bank of 30 high-pass filters. These filters highlight statistical irregularities that occur when an image is manipulated.



Hand-Crafted Feature Extraction

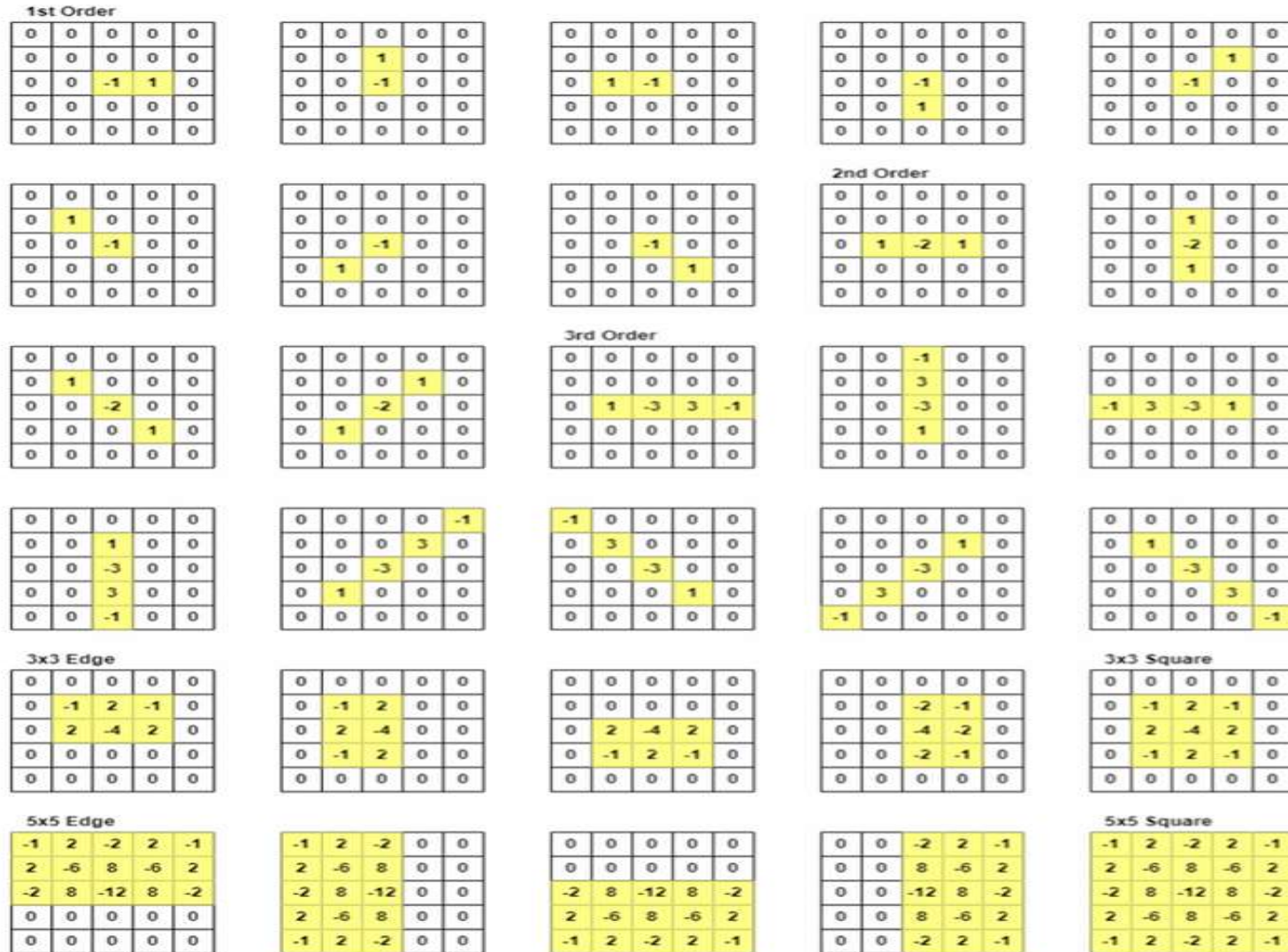


Fig: 30 high pass SRM filters

Hand-Crafted Feature Extraction

□ Noise Residuals from Spatial Rich Model (SRM)

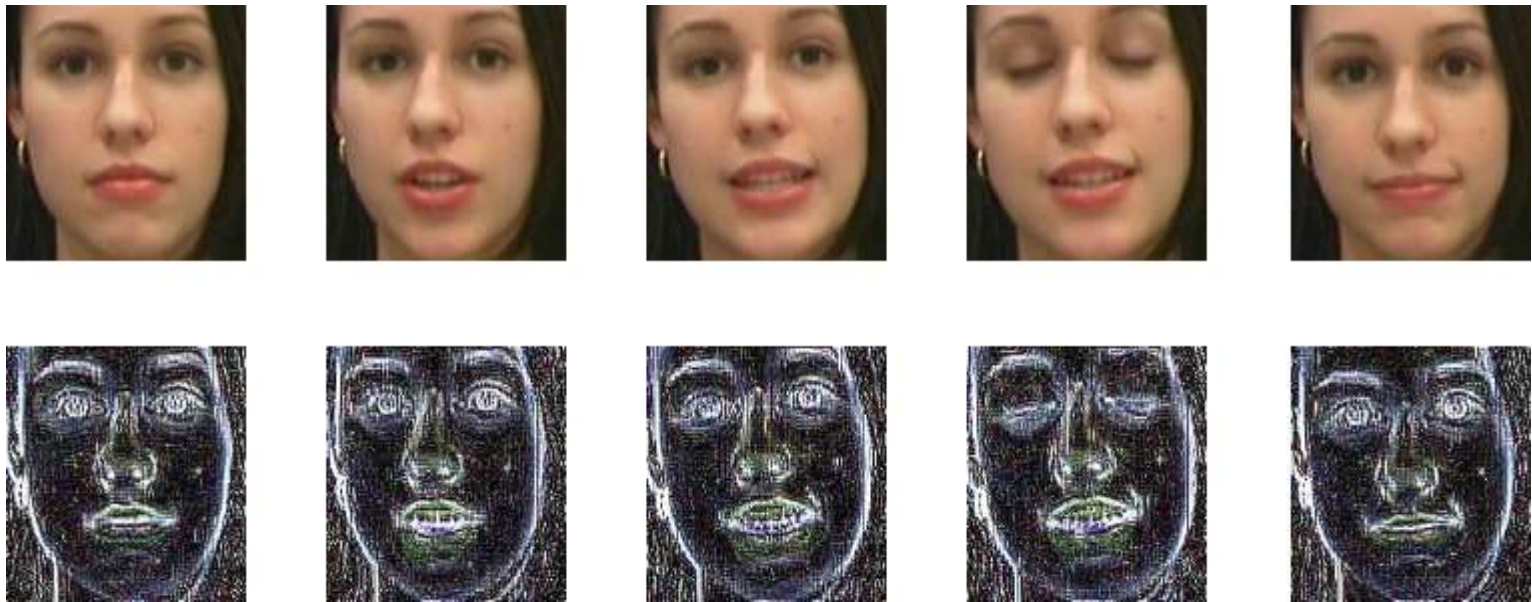
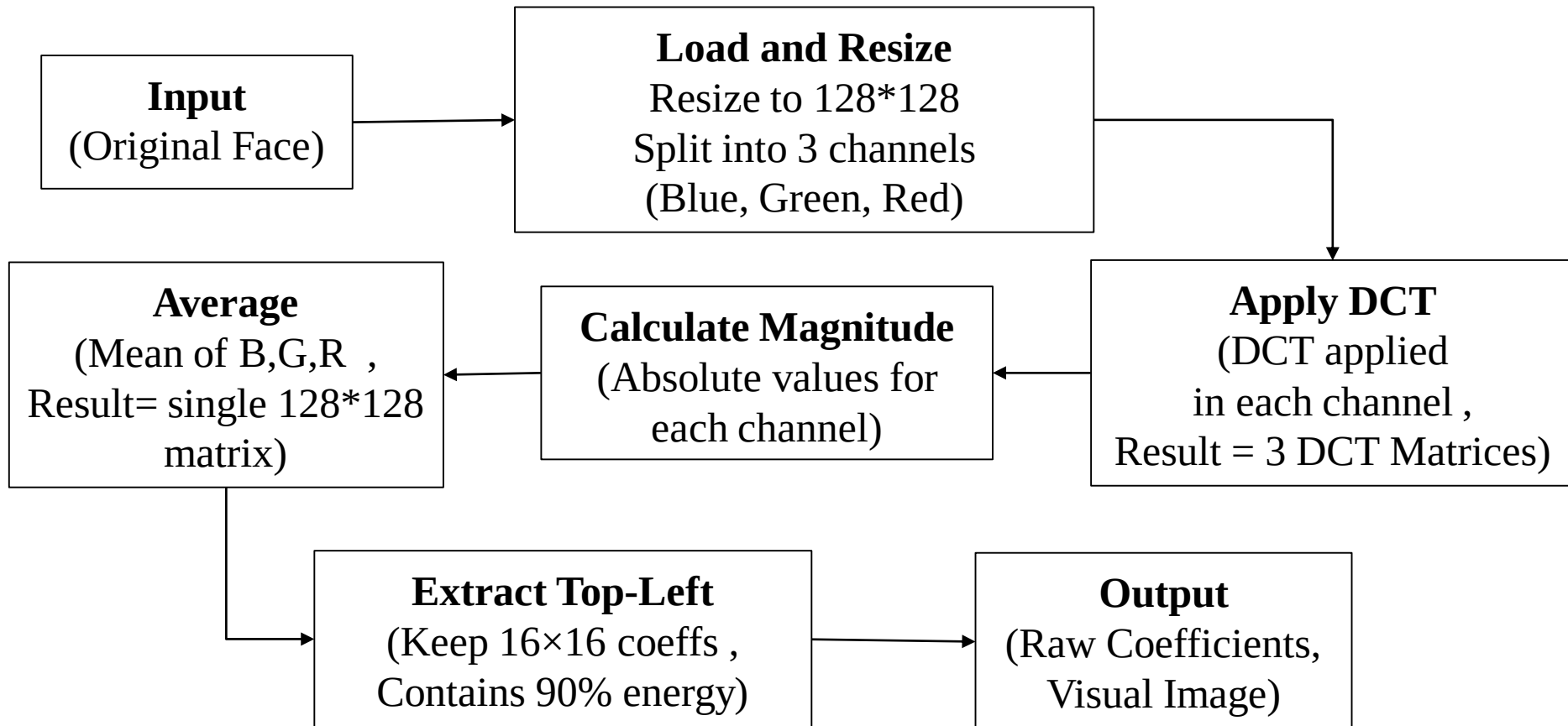


Fig: Authentic Image and Its SRM

Hand-Crafted Feature Extraction

❑ Discrete Cosine Transform (DCT)

DCT examines the frequency components of an image, making it effective for detecting deepfake artifacts.



Hand-Crafted Feature Extraction

□ Discrete Cosine Transform (DCT)

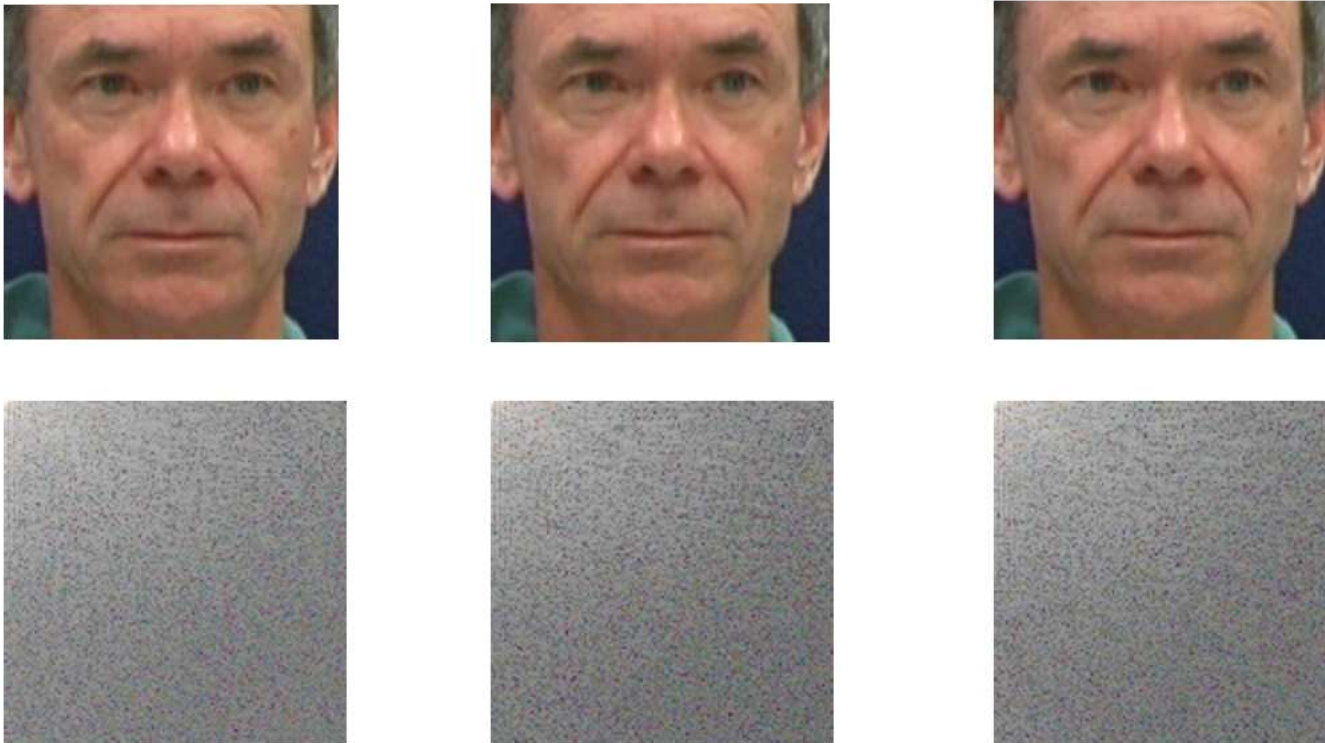
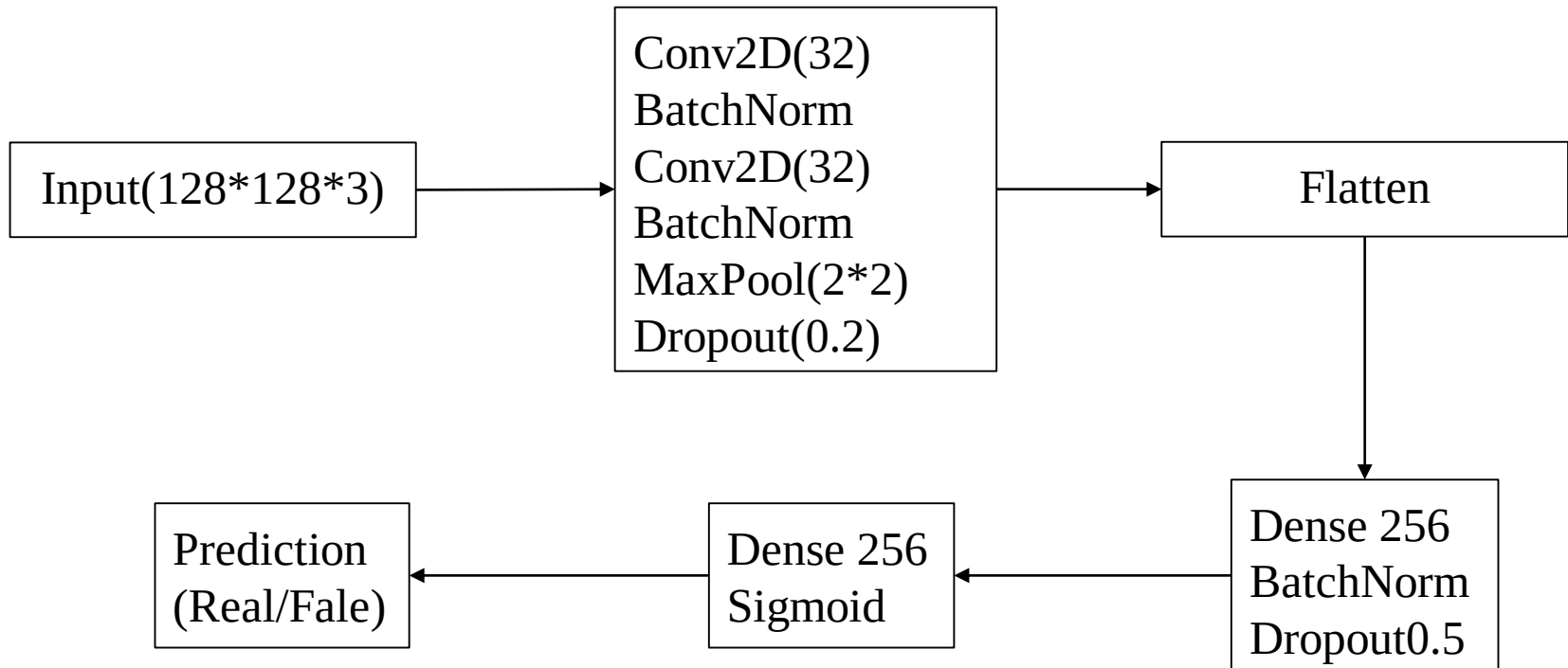


Fig: Authentic Image and Its DCT

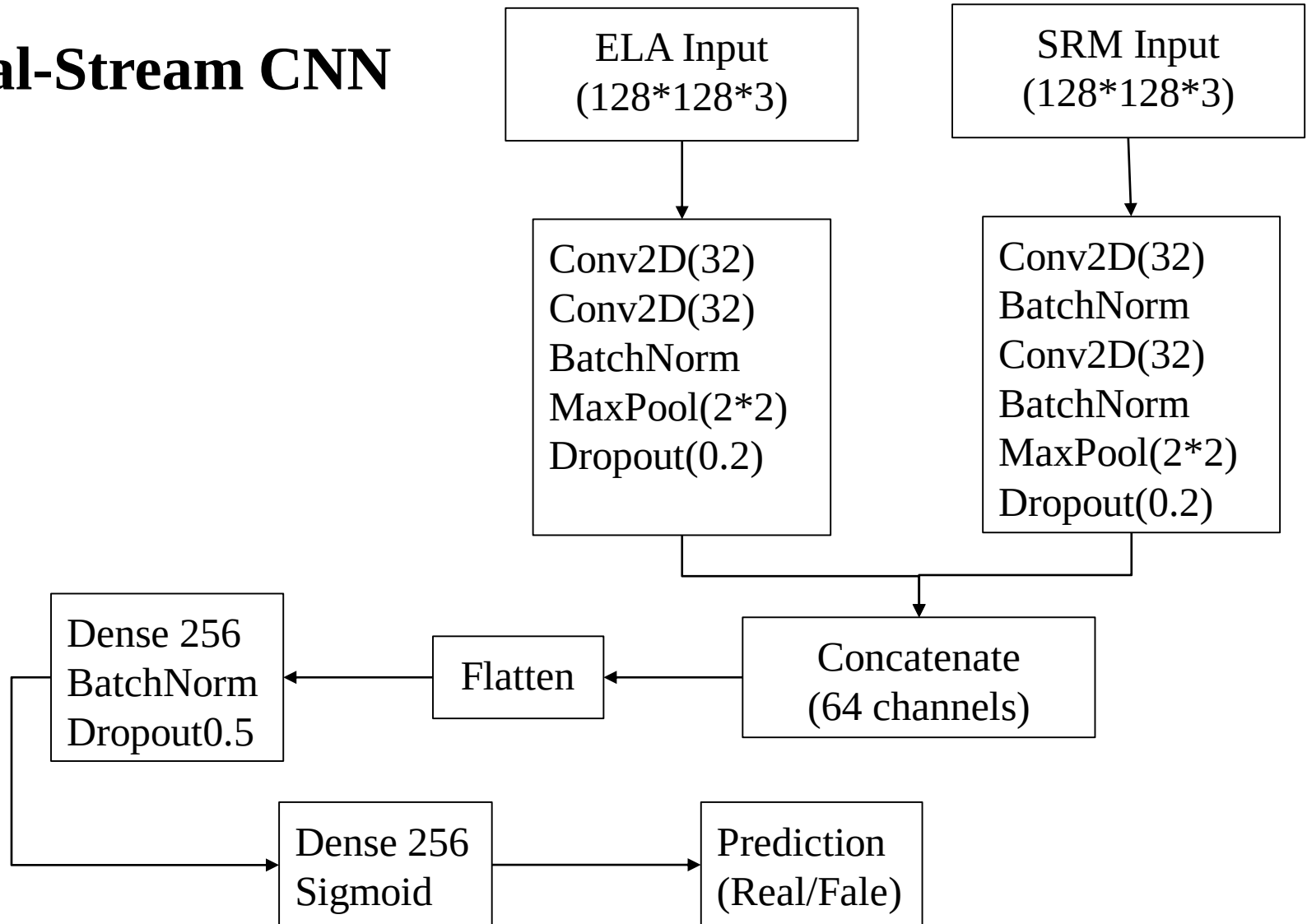
Deep Learning Architecture

□ Single-Stream CNN



Deep Learning Architecture

❑ Dual-Stream CNN





Results

Dataset	Method	Accuracy	AUC
FF-FS	ELA-SRM	98.12	0.9985
FF-FS	DCT-ELA	97.15	0.9958
FF-FS	DCT-SRM	97.77	0.9981
FF-FS	DCT	91.61	0.971
FF-FS	ELA	94.95	0.9875
FF-FS	SRM	99.78	0.9995



Thank You !